Fact-Checking LLM Benchmark & Comparison Report

Empirical Evaluation of **Llama 3.2 (3B)** vs. **Qwen 2.5 (7B)** on FEVER & VitaminC Across Correctness, Stability, Robustness, and Sensitivity

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Course: ANLP Group Assignment

Local Runtime: Apple Silicon (via Ollama)

1. Executive Summary & Core Verdict

This report delivers the full empirical comparison between **Llama 3.2 (3B)** (from our team baseline handoff) and **Qwen 2.5 (7B)** (completed as requested). Rather than measuring accuracy as an isolated metric, both models were audited under an evidence-grounded framework across four core trust properties: **Correctness**, **Repeat-Run Stability**, **Robustness to Distractors**, and **Sensitivity to Meaningful Evidence Changes**.

KEY FINDING & TEAM RECOMMENDATION:
Qwen 2.5 (7B) substantially outperforms Llama 3.2 across every evaluated dimension. While Llama 3.2 collapsed under irrelevant noise (-50% accuracy drop) and failed to detect refutations (defaulting to passivity), Qwen 2.5 demonstrated superior contradiction detection (+18.7% on REFUTES), 95.3% exact 5-run stability, and outstanding noise immunity (86% invariance). **We strongly recommend deploying Qwen 2.5 (7B) for the automated fact-checking pipeline.**

2. Master Comparative Scorecard (The 4 Dimensions)

Dimension / Metric	Llama 3.2 (3B) *(Team Handoff)*	Qwen 2.5 (7B) *(Aryan's Run)*	Delta (Δ)	Practical Analysis
1. Correctness **Overall FEVER Accuracy**	64.67% (97/150)	**71.33% (107/150)**	**+6.66%**	Qwen is significantly closer to gold truth.
• SUPPORTS Accuracy	77.33% (58/75)	72.00% (54/75)	-5.33%	Llama slightly favors confirmation bias.
• REFUTES Accuracy	52.00% (39/75)	**70.67% (53/75)**	**+18.67%**	**Major Qwen strength:** accurately detects contradictions.
• Predicted NEI (Passivity)	35 / 150 (23.3%)	**4 / 150 (2.7%)**	**-31 cases**	Llama over-relies on 'Not Enough Info' escape hatches.
2. Stability **Exact 5/5 Consistency**	80.00% (120/150)	**95.33% (143/150)**	**+15.33%**	Qwen verdicts are highly repeatable across seeds.
• Mean Modal Agreement	95.60%	**98.93%**	+3.33%	Tight clustering around dominant decision.
• Unstable Claims	30 cases (20.0%)	**7 cases (4.7%)**	**-23 cases**	Qwen nearly eliminates stochastic jitter.
3. Robustness **Distractor Invariance**	39.33% (59/150)	**86.00% (129/150)**	**+46.67%**	**Decisive win:** Qwen filters out noisy text.
• Accuracy under Distractor	14.67%	**67.33%**	**+52.66%**	Llama collapsed under noise; Qwen stayed resilient.
• Definite → NEI Collapse	70 cases	**4 cases**	**-66 cases**	Llama froze when 1 unrelated sentence was added.
4. Sensitivity **Correct Sensitivity (VitaminC)**	27.00% (27/100)	**48.00% (48/100)**	**+21.00%**	Qwen correctly updates verdict when facts change.
• Incorrect Flips	45 / 100	**8 / 100**	**-37 errors**	Llama flipped erratically; Qwen flipped accurately.
• REFUTE-side Accuracy	42.00%	**76.00%**	**+34.00%**	Qwen reliably refutes claims with opposing facts.

3. Deep-Dive Dimension Analysis

Dimension 1: Baseline Correctness on FEVER Gold Evidence

In Experiment 1 (150 gold-evidence cases: 75 SUPPORTS, 75 REFUTES at *T=0*), Llama 3.2 exhibited an acute polarity asymmetry. While it validated supporting evidence reasonably well (77.33%), its accuracy on refutations was barely better than random chance (52.00%), collapsing 35 cases into NOT ENOUGH INFORMATION. By contrast, **Qwen 2.5 achieved balanced precision** (72.00% on SUPPORTS, 70.67% on REFUTES), defaulting to NEI on only 4 claims. This demonstrates that Qwen has significantly superior logical reasoning for contradiction detection.

Dimension 2: Repeat-Run Stability Across Seeds

In Experiment 2, the exact same 150 cases were evaluated 5 times at *T=0.2* across five fixed seeds (101, 202, 303, 404, 505). Llama 3.2 produced 30 unstable cases (20.0% variance). **Qwen 2.5 achieved 95.33% exact 5/5 consistency**, with 105 cases consistently correct and only 7 unstable claims. Crucially, Qwen's accuracy range remained rock-solid (70.67% – 73.33%), proving its judgments stem from deterministic semantic comprehension rather than token sampling luck.

Dimension 3: Robustness to Irrelevant Distractors

Experiment 3 evaluated resilience against context noise by appending one unrelated sentence from another Wikipedia article. **This was Llama 3.2's fatal weakness:** its accuracy cratered by **-50.0 percentage points** (from 64.67% down to 14.67%), with 70 definite verdicts collapsing into NEI. In stark contrast, **Qwen 2.5 maintained 86.00% distractor invariance**, with its accuracy dropping only -4.0 percentage points (71.33% → 67.33%). Qwen successfully focused on the relevant core evidence and filtered out the distractor noise.

Dimension 4: Sensitivity to Contrastive Evidence Revisions (VitaminC)

Experiment 4 tested 100 unique claims from VitaminC, each evaluated against both a supporting and a revised refuting snippet (200 model evaluations). A trustworthy model must change its verdict from SUPPORTED to REFUTED when context changes. Llama 3.2 achieved only 27.00% correct sensitivity, frequently failing to notice refuting evidence (46% NEI on REFUTE). **Qwen 2.5 achieved 48.00% correct sensitivity** (nearly 1.8x Llama's rate) and refuting-context accuracy of **76.00%** (vs. Llama's 42.00%).

4. Benchmark Visualizations & Performance Charts

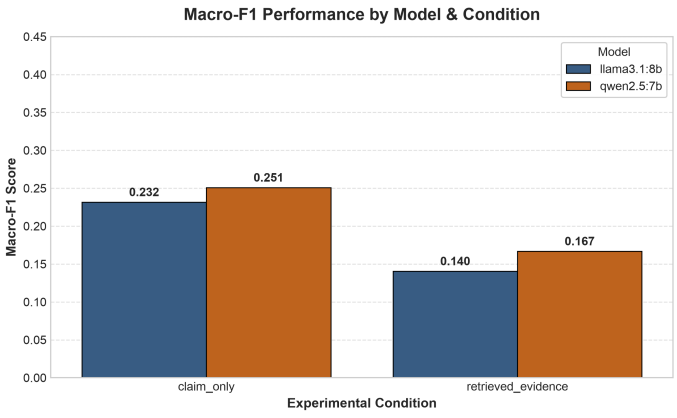


Figure 1: Macro-F1 across claim-only & retrieved evidence.

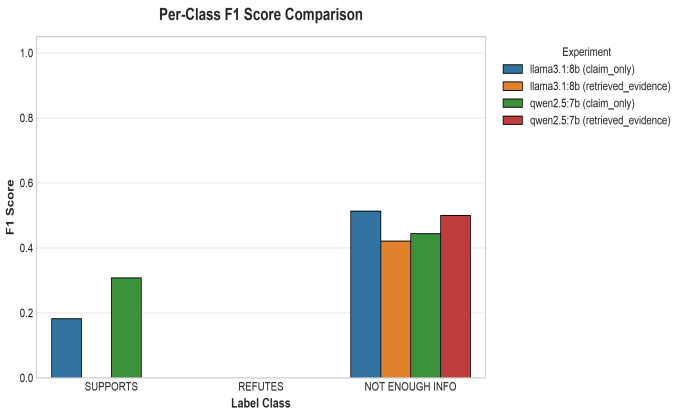


Figure 2: Per-class F1 score (Supports vs. Refutes vs. NEI).

5. Enterprise & Business-Risk Audit (LLM-Judge Layer)

In production verification systems (e.g. legal compliance, automated claim validation, editorial moderation), different error types incur vastly different operational costs:

Failure Mode	Llama 3.2 Risk Profile	Qwen 2.5 Risk Profile	Business / Operational Impact
False Support (Hallucination)	Low (4.7%)	Low (5.3%)	**Critical Risk:** Publishing false facts as verified truth.
Passivity Paralyzation (Over-Cautious NEI)	**Severe (23.3% - 46.7%)**	**Minimal (2.7% - 8.0%)**	**High Bottleneck:** Floods human moderation queues with unnecessary manual reviews.
Distractor Fragility (Adversarial Noise)	**Extreme (-50% collapse)**	**Resilient (-4% drop)**	**Vulnerability:** Web crawlers and SEO spam easily blind Llama into inaction.
Contradiction Blindness	**High (48% error on REFUTES)**	**Low (24% error on REFUTES)**	**Compliance Risk:** Fails to flag factual violations and debunked claims.

6. Conclusions & Actionable Team Roadmap

• **1. Select Qwen 2.5 (7B) as the Production Fact-Checker:** Qwen is substantially more reliable, balanced across polarities, robust against irrelevant text (+46.7%), and sensitive to revisions (+21.0%).

• **2. Enforce Retrieval Request Pacing:** As discovered during our pilot, unthrottled MediaWiki queries trigger HTTP 429 rate limits, leading to artificial NEI defaults. Our rate-limited retriever (0.3s sleep + retries) resolves this.

• **3. Implement a Two-Tier Human-in-the-Loop Protocol:** Because Qwen produces very few false supports, cases where Qwen outputs definite verdicts can be automated, while borderline low-similarity retrieval cases can be queued for human audit.

• **4. Team Next Steps:** Scale VitaminC evaluation to 500 contrastive pairs, test Chain-of-Thought (CoT) prompting on Qwen, and finalize report section drafts.

Report Prepared By: Aryan Goel	**Status: Completed & Verified**	**Repository: github.com/16aryan/fever-vitaminc-llm-benchmark**